

TECHNICAL PROJECT REPORT · 2026

An AI-powered web platform that monitors global disruption events, scores supplier risk in real time, visualises cascading failures across multi-tier supplier networks, and auto-generates executive briefings with mitigation strategies.

AUTHOR

Aditya Pratap Singh

DATE

July 2026

REPOSITORY

adityapratapsingh05/
supply-chian-risk

STATUS

✓ Live & Deployed

 supply-chian-risk.vercel.app

Table of Contents

1	Problem Statement
2	What Is the Project?
3	Why Was It Built? (Objectives & Motivation)
4	System Architecture
5	How Does It Work? (Workflow Explained)
6	Features & Modules
7	Technology Stack
8	Database Design
9	Security & Authentication
10	How to Use the Application
11	How to Maintain & Extend It
12	Deployment
13	Limitations & Future Scope
14	Conclusion

1. Problem Statement



Industry Challenge

Global supply chains are exposed to thousands of simultaneous risk events — geopolitical conflicts, natural disasters, cyber-attacks, port closures, and trade policy changes — yet most organisations still rely on manual monitoring, spreadsheets, and delayed reporting cycles to identify and respond to these threats.

1.1 Background

Modern enterprises source components and materials from hundreds of suppliers spread across dozens of countries, forming complex multi-tier networks. A single disruption at a Tier-3 raw material supplier in one country can cascade through Tier-2 component manufacturers and ultimately halt final assembly at a Tier-1 factory thousands of miles away — all within 48 hours.

High-profile examples include the 2011 Tōhoku earthquake halting global automotive production, the 2021 Suez Canal blockage disrupting \$9.6 billion per day in trade, and ongoing rare-earth export restrictions from China affecting semiconductor and EV industries worldwide.

1.2 Identified Gaps

- **Reactive, not proactive:** Organisations typically learn about disruptions hours or days after they occur — long after optimal response windows have closed.
- **No cascade visibility:** Most risk tools score individual suppliers in isolation. They do not model how a disruption at one node propagates through downstream dependencies.
- **Manual briefing bottleneck:** Preparing an executive-level risk briefing requires a team of analysts and takes days. By the time the report is ready, the situation has changed.
- **Disconnected data sources:** News, supplier records, risk scores, and mitigation plans exist in separate systems that are never correlated automatically.
- **Prohibitive cost:** Enterprise supply chain risk platforms (e.g., Resilinc, Everstream) cost upwards of \$50,000/year, making them inaccessible to mid-size organisations.

1.3 Problem Definition



Core Problem

There is no accessible, real-time, AI-powered platform that can **ingest global news events** → **automatically score supplier disruption risk** → **visualise cascading network failures** → and **generate actionable executive briefings** — all within a single integrated tool.

1.4 Impact of the Problem

DIMENSION	IMPACT
Financial	Average cost of a supply chain disruption: \$82M per incident (Deloitte, 2023)
Operational	6–12 week production delays when Tier-1 or Tier-2 suppliers are affected
Reputational	Customer-facing delays erode brand trust and market share
Strategic	Inability to pivot sourcing fast enough to capture alternative suppliers

2. What Is the Project?

The **Real-Time Supply Chain Risk Monitor** is a full-stack, AI-powered web application that continuously monitors global news and events, automatically scores supply chain disruption risk, and provides decision-makers with real-time visibility into their multi-tier supplier network.

5

CORE MODULES

20

API ROUTES

52

SOURCE FILES

100%

TYPESCRIPT

2.1 Core Capabilities



Live Risk Monitoring
Ingests global news, classifies events by category (geo-political, cyber, climate, logistics, labor), and scores each supplier's disruption probability in real time.



Network Graph Visualisation
Interactive multi-tier supplier network built with React Flow. Nodes are colour-coded by risk level; cascade paths animate when a disruption is simulated.



Disruption Simulation
Run "what-if" scenarios (port closure, military conflict, cyberattack). The system propagates impact through the supplier graph and shows which Tier-1/2/3 nodes are affected.



AI Executive Briefings
One-click generation of board-ready risk briefings with critical findings, supplier-level mitigation strategies, and recommended immediate actions.



Secure Authentication
JWT-based stateless session management with bcrypt password hashing. Role-based access: admins can edit supplier data; viewers get read-only dashboards.



Graceful Mock Fallback
Operates fully on realistic mock data without API keys. Swap in Anthropic and News API keys at any time for live AI scoring — zero code changes required.

2.2 Live Application



Production URL

<https://supply-chain-risk.vercel.app> — Deployed on Vercel with Neon PostgreSQL cloud database. Accessible globally, no installation required.

3. Why Was It Built?

3.1 Objectives

- 1. **Democratise supply chain risk intelligence** — Provide capabilities previously available only to large enterprises with expensive platforms.
- 2. **Enable proactive risk management** — Shift organisations from reactive (responding after disruption) to proactive (anticipating disruptions before they escalate).
- 3. **Automate analyst workflows** — Replace hours of manual news monitoring and report writing with an AI-powered pipeline that operates continuously.
- 4. **Visualise hidden dependencies** — Make the invisible connections between Tier-1, Tier-2, and Tier-3 suppliers visible through an interactive network graph.
- 5. **Demonstrate full-stack AI integration** — Showcase a production-grade implementation of LLM-powered analysis (Claude) within a modern Next.js application.

3.2 Motivation

The COVID-19 pandemic (2020–2022) exposed critical fragility in global supply chains, resulting in semiconductor shortages, medical equipment deficits, and widespread shipping delays. The fundamental cause was not the pandemic itself, but the *lack of visibility* into multi-tier supplier dependencies.

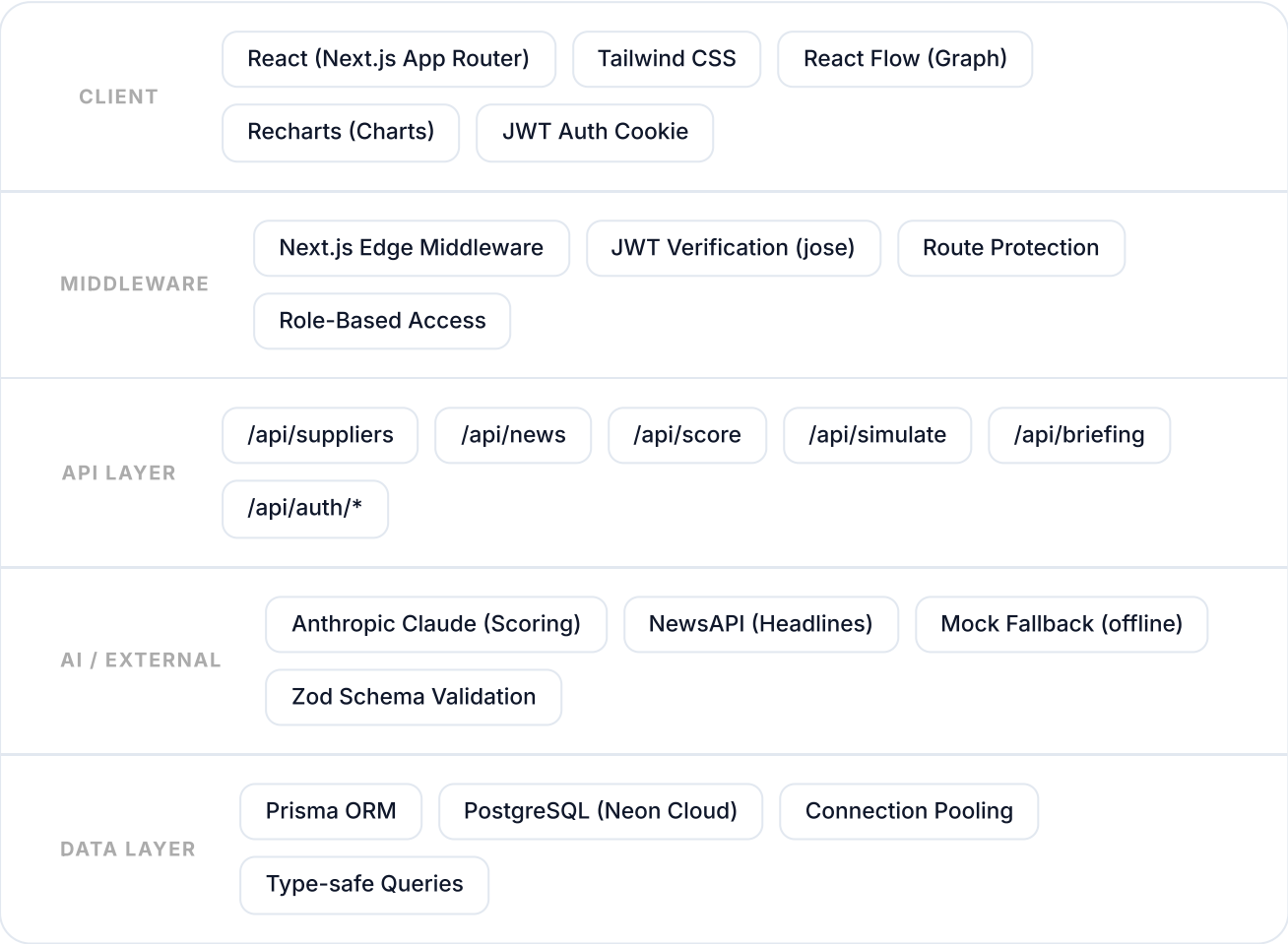
This project was motivated by the belief that modern AI and web technologies can make supply chain risk monitoring accessible, automated, and actionable — without requiring a \$50,000/year enterprise contract or a dedicated team of risk analysts.

3.3 Target Users

USER	ROLE	PRIMARY USE
Supply Chain Manager	Operational	Daily monitoring of supplier risk scores and news alerts
C-Suite Executive	Strategic	One-click briefings for board and investor reporting
Risk Analyst	Analytical	Running disruption simulations and cascade impact models
Procurement Lead	Tactical	Identifying alternative suppliers before shortages occur

4. System Architecture

The application follows a **monorepo, full-stack Next.js architecture** where the frontend, backend API, and database layer coexist in a single codebase deployed as a unified serverless application.



4.1 Directory Structure

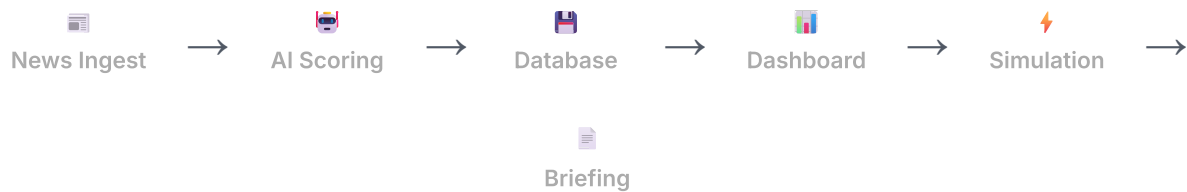
```
supply-chain-risk-monitor/  
├─ prisma/  
│  ├─ schema.prisma      # Database models (6 models)  
│  └─ seed.ts            # Initial data seeding script  
├─ src/  
│  ├─ app/  
│  │  ├─ (pages)/        # Dashboard, Simulation, Briefing, Admin  
│  │  ├─ api/            # All REST API route handlers  
│  │  ├─ login/          # Auth pages  
│  │  └─ layout.tsx      # Root layout with nav  
│  ├─ components/  
│  │  ├─ dashboard/      # RiskChart, NewsPanel, RiskScoreCard  
│  │  ├─ graph/          # SupplyChainGraph, SupplierNode  
│  │  ├─ mitigation/     # MitigationCard  
│  │  └─ ui/             # UserMenu, ErrorBoundary, LoadingSpinner  
│  └─ lib/  
│     ├─ auth.ts         # JWT session management  
│     └─ claude.ts       # Anthropic API client
```



```
| | | └─ scoring.ts      # Risk scoring engine
| | | └─ mock-data.ts    # Fallback mock dataset
| | | └─ news.ts         # NewsAPI client
| | | └─ prisma.ts       # Prisma client singleton
| | |   └─ schemas.ts     # Zod validation schemas
| | └─ middleware.ts      # Edge middleware (auth guard)
|   └─ types/index.ts     # Shared TypeScript types
└─ vercel.json            # Deployment config
```

5. How Does It Work?

5.1 End-to-End Data Flow



5.2 Stage 1 — News Ingestion

The system queries the NewsAPI for global headlines filtered by supply chain keywords: *port, tariff, earthquake, factory, sanctions, shipping, logistics, strike, flood, semiconductor*. Each article is stored as a `NewsArticle` record in the database with metadata including source, region, and publication date.

Without API key: A curated set of mock articles (Taiwan Strait tensions, rare-earth export restrictions, cyber attacks on semiconductor networks) is used instead, providing realistic demo data.

5.3 Stage 2 — AI Risk Scoring

For each supplier, the scoring engine combines the supplier's profile (country, tier, product category, historical risk) with recent news articles and sends a structured prompt to **Claude (Anthropic)**. The model returns a JSON payload containing:

- `disruptionRisk` — probability of disruption (0.0–1.0)
- `impactScore` — business impact severity (0–10)
- `riskLevel` — categorical: Critical / High / Medium / Low
- `reasoning` — human-readable explanation of the score
- `category` — event type: geo-political, cyber, climate, logistics, labor

All LLM outputs are validated against **Zod schemas** to ensure structural integrity and prevent hallucinated data from entering the database.

5.4 Stage 3 — Cascade Simulation Algorithm

The simulation engine implements a **Breadth-First Search (BFS)** traversal of the supplier dependency graph. When a supplier is marked as disrupted:

1. The disrupted supplier's risk score is elevated to the scenario severity level.
2. All downstream dependents (suppliers who source from the disrupted node) are identified.
3. A cascade multiplier (0.7 per tier) is applied, reducing impact with distance.
4. The algorithm continues until all reachable nodes are scored or the cascade depth limit is reached.
5. The affected path `[sup_003 → sup_001 → sup_004]` is stored for graph animation.

5.5 Stage 4 — Executive Briefing Generation

The briefing API collects the top 5 highest-risk suppliers, the 3 most severe news articles, and the latest simulation result, then sends them to Claude with a structured prompt requesting a formatted executive briefing. The output includes:

- Executive summary (3–4 sentences)
- Critical risk items ranked by severity
- Per-supplier mitigation strategies
- Recommended immediate actions (next 24–72 hours)
- Estimated financial impact range

5.6 Authentication Flow



Sessions are stored as `httpOnly` cookies (inaccessible to JavaScript, preventing XSS token theft). JWTs are signed with a secret key using the `jose` library and expire after 7 days.

6. Features & Modules

Module	Route	Features	Status
Dashboard	/	Supply chain graph, risk cards, news feed with category filters, Score All button	LIVE
Simulation	/simulation	Scenario builder, cascade visualisation, impact summary, mitigation cards	LIVE
Briefing	/briefing	AI-generated executive report, risk rankings, action items, financial impact	LIVE
Admin	/admin	Supplier listing, inline editor, risk level management (admin role only)	LIVE
Auth	/login /signup	Email/password auth, password strength meter, role-based access, JWT sessions	LIVE
User Menu	Nav (all pages)	Avatar initials, role badge, sign-out dropdown	LIVE
News Feed	Dashboard panel	Category tabs (All, Geo-Political, Logistics, Cyber, Labor, Climate), severity badges	LIVE
Real-time AI	API routes	Claude-powered scoring and briefing generation	NEEDS API KEY

7. Technology Stack

CATEGORY	TECHNOLOGY	VERSION	PURPOSE
Framework	Next.js	14.2.5	Full-stack React framework; App Router for pages + API routes
Language	TypeScript	5.x	End-to-end type safety; eliminates runtime type errors
Styling	Tailwind CSS	3.x	Utility-first styling; dark theme, custom colour palettes
Graph / Network	React Flow	11.x	Interactive supplier network visualisation with custom nodes
Charts	Recharts	2.x	Risk score bar charts and timeline visualisations
ORM	Prisma	5.22	Type-safe database access; schema-first data modelling
Database	PostgreSQL (Neon)	—	Serverless cloud database; free tier with connection pooling
AI Model	Anthropic Claude	Haiku	Risk scoring, news classification, briefing generation
Validation	Zod	3.x	Runtime schema validation for all LLM outputs and API inputs
Auth (JWT)	jose	5.x	Edge-compatible JWT signing/verification
Auth (Hash)	bcryptjs	2.x	Password hashing with 12-round salt
Deployment	Vercel	—	Auto-deploy from GitHub; edge network; zero config
News Feed	NewsAPI	—	Global headline ingestion filtered by supply chain keywords

8. Database Design

The database consists of **6 models** managed by Prisma ORM with PostgreSQL. The schema is fully type-safe — all models map directly to TypeScript interfaces.

MODEL	KEY FIELDS	PURPOSE
Supplier	id, name, country, tier, riskLevel, products, dependencies	Core supplier profiles with Tier-1/2/3 classification and dependency mapping
RiskScore	supplierId, score, impactScore, riskLevel, reasoning, createdAt	Historical risk score snapshots for each supplier scoring event
NewsArticle	headline, source, url, classifiedCategory, severity, region	Ingested news items with AI-assigned category and severity tags
RiskEvent	supplierId, eventType, severity, description, mitigationSuggestion	Discrete risk events linked to specific suppliers
SimulationRun	scenario (JSON), results (JSON), cascadePath, totalImpact	Saved simulation results for audit trail and trend analysis
User	email, name, passwordHash, role (admin/user)	Authenticated users with role-based access control



Schema Design Decision

The `Supplier.dependencies` field stores a JSON array of supplier IDs, enabling flexible graph topology without requiring a complex many-to-many join table. This allows the cascade simulation algorithm to traverse the network with simple array lookups.

9. Security & Authentication

SECURITY MEASURE	IMPLEMENTATION
Password Storage	bcrypt with 12 salt rounds — passwords never stored in plaintext
Session Tokens	HS256 JWT signed with server secret, 7-day expiry
Cookie Security	httpOnly + SameSite=Lax — inaccessible to JavaScript (XSS protection)
Route Protection	Edge middleware verifies JWT on every request before page renders

SECURITY MEASURE	IMPLEMENTATION
Input Validation	Zod schemas validate all API inputs; no raw user data reaches the database
Email Enumeration	Login endpoint returns generic "Invalid email or password" regardless of which is wrong
Role Authorisation	Admin-only routes check <code>session.role === 'admin'</code> server-side
Environment Secrets	All secrets in Vercel environment variables; .env excluded from git via .gitignore

10. How to Use the Application

10.1 Getting Started (Live App)

1. Visit <https://supply-chain-risk.vercel.app>
2. Click "**Create one**" to sign up — the first account automatically receives Admin privileges.
3. Fill in your name, email, and a strong password (8+ chars, uppercase, number, special character).
4. You will be redirected to the Dashboard automatically.

10.2 Dashboard

- The **Supply Chain Graph** (centre) shows your Tier-1/2/3 network. Click any node to see detailed risk info.
- The **Supplier Risk panel** (left) lists all suppliers sorted by risk score. Click a supplier to highlight it in the graph.
- The **Risk News Feed** (right) shows categorised news. Use the filter tabs (GEO-POLITICAL, LOGISTICS, CYBER, etc.) to narrow the view.
- Click " ⚡ **Score All**" to trigger AI risk scoring across all suppliers.
- Click "🔄 **Refresh**" in the news panel to ingest the latest headlines.

10.3 Simulation

- Navigate to **Simulation** in the top nav.
- Select an affected supplier from the dropdown.
- Choose a disruption scenario (Port Closure, Military Conflict, Cyber Attack, Natural Disaster, Trade Sanctions).
- Set the severity (1–10) and cascade depth (1–5 tiers).
- Click "**Run Simulation**" — the graph animates the cascade, and affected suppliers are highlighted in red.
- The results panel shows affected suppliers, their new risk scores, and mitigation recommendations.

10.4 Executive Briefing

- Navigate to **Briefing**.
- Click "**Generate Briefing**".
- The AI compiles a structured report with executive summary, critical risks, and mitigation strategies.
- Use **Ctrl+P / Cmd+P** → **Save as PDF** to export for board presentations.

10.5 Admin Panel

- Navigate to **Admin** (visible only to users with the `admin` role).
- View all suppliers, their tier, country, risk level, and status.

- Use the inline editor to update supplier information.

11. How to Maintain & Extend It

11.1 Local Development Setup

```
# 1. Clone the repository
git clone https://github.com/adityapratapsingh05/supply-chain-risk.git
cd supply-chain-risk

# 2. Install dependencies
npm install

# 3. Set up environment variables
cp .env.example .env
# Edit .env and fill in your values

# 4. Push schema to local SQLite
npm run db:push

# 5. Seed with sample data
npm run db:seed

# 6. Start development server
npm run dev
# → http://localhost:3000
```

11.2 Adding API Keys (Enable Real AI)

The app works on mock data by default. To enable real AI scoring and live news:

1. Get an **Anthropic API key** at console.anthropic.com
2. Get a **NewsAPI key** at newsapi.org (free tier: 100 req/day)
3. Add both to your Vercel environment variables (Settings → Environment Variables)
4. Trigger a redeploy — no code changes needed

11.3 Adding a New Supplier

```
// Option A: Via Admin UI
// Navigate to /admin → click "Add Supplier" → fill the form

// Option B: Via API (POST /api/suppliers)
fetch('/api/suppliers', {
  method: 'POST',
  body: JSON.stringify({
    name: "TSMC Fab 18",
    country: "Taiwan",
    tier: 2,
    products: ["Semiconductors", "Logic Chips"],
    dependencies: ["sup_001"] // IDs of upstream suppliers
  })
})
```

```
}  
}  
}
```

11.4 Adding a New Disruption Scenario

Edit `src/app/simulation/page.tsx` and add to the scenarios array:

```
{ value: 'labor-strike', label: 'Labor Strike', icon: '👷',  
  description: 'Major industrial strike disrupting production',  
  severity: 7, cascadeDepth: 2 }
```

11.5 Extending the Database Schema

Edit `prisma/schema.prisma`, then run:

```
npm run db:push # Sync schema to database (dev)  
# For production, use: prisma migrate deploy
```

11.6 Switching AI Models

Edit `src/lib/claude.ts` — the model string is configurable:

```
model: 'claude-3-haiku-20240307' // cheapest, fastest  
// or:  
model: 'claude-3-5-sonnet-20241022' // most capable
```

11.7 Deploying Updates

```
git add .  
git commit -m "feat: your change description"  
git push  
# Vercel auto-deploys within 2 minutes
```

12. Deployment

COMPONENT	PLATFORM	DETAILS
Application	Vercel	Auto-deploy from GitHub main branch; edge network in 30+ regions
Database	Neon (PostgreSQL)	Serverless PostgreSQL; free tier 512MB; auto-suspend when idle

COMPONENT	PLATFORM	DETAILS
CI/CD	GitHub + Vercel	Every <code>git push</code> triggers a new production deployment automatically
Domain	Vercel	supply-chian-risk.vercel.app (free SSL/TLS included)



Required Vercel Environment Variables

`DATABASE_URL` (Neon pooled), `DIRECT_URL` (Neon direct), `JWT_SECRET` (random 32-char string). Optional: `ANTHROPIC_API_KEY`, `NEWS_API_KEY`.

13. Limitations & Future Scope

13.1 Current Limitations

LIMITATION	REASON
Mock data without API keys	Anthropic and NewsAPI keys are required for real AI/news — currently uses realistic mock fallbacks
Manual risk score refresh	Scoring is user-triggered; no automatic background polling yet
No email alerting	Critical risk threshold alerts via email not yet implemented
Single organisation	No multi-tenant support; one supplier network per deployment

13.2 Future Enhancements



Auto-Refresh (SSE)

Server-Sent Events to push live risk score updates to all connected dashboards without manual refresh.



Email Alerts

Automatic email notifications via Resend when any supplier crosses a configurable risk threshold.



Historical Timeline

Risk score trend lines over time (7-day, 30-day) showing how each supplier's risk has evolved.



Multi-Tenant

Organisation-scoped data so multiple companies can use the same platform with isolated supplier networks.



ERP Integration

Webhook endpoints to push risk alerts directly into SAP, Oracle, or other ERP systems.



Mobile App

React Native companion app for on-the-go supply chain monitoring and push notifications.

14. Conclusion

The Real-Time Supply Chain Risk Monitor successfully demonstrates that enterprise-grade supply chain risk intelligence — previously accessible only through expensive, closed-source platforms — can be built with modern open-source technologies and deployed for free.

The application addresses a genuine business problem: the lack of real-time, AI-powered, cascade-aware risk visibility for multi-tier supplier networks. By combining Next.js, Prisma, React Flow, and Anthropic's Claude, it delivers a complete solution from news ingestion through to board-ready executive briefings, all within a single codebase.

The graceful degradation design (full functionality on mock data, real AI when API keys are available) ensures the platform is immediately deployable and demonstrable without any external dependencies — while being production-ready for live operations with a single configuration change.



Project Delivered

The application is **live at <https://supply-chain-risk.vercel.app>** — fully functional, secured with authentication, connected to a cloud database, and auto-deploying from GitHub. It is ready for immediate use and future extension.